One-Channel Mixed-Signal Impedance Front End

Project	Senior Design: Low-Power Agricultural Impedance Sensing
Architecture	Fixed-gain TIA, 8-bit SAR ADC, and minimal SPI backend
Team	ASIC: Krishna, Taara, and Vidhu; systems software: three CS team members
Revision	2.1 – Simplified ASIC and Software-System Baseline
Date	September 1, 2026
Status	Simplified implementation specification; circuit values remain subject to simulation

This revision reduces the original four-channel autonomous measurement system to one externally excited channel. The ASIC retains a meaningful mixed-signal contribution while moving waveform generation, synchronous demodulation, averaging, and calibration to the MCU.

Document Purpose

This document defines a reduced-scope ASIC that can be completed and verified by a three-person ASIC team, with three additional CS team members building the software system above the MCU. The chip senses one electrode current with a transimpedance amplifier (TIA), digitizes the TIA output with an 8-bit successive-approximation (SAR) ADC, stores the result, and returns it to an external microcontroller through SPI. The MCU and board generate the excitation waveform and perform phase-sensitive processing in software.

The primary source remains the submitted proposal, *Mixed-Signal Impedance Sensing ASIC for Battery-Free Agricultural IoT Nodes*. The proposal establishes the agricultural sensing use case, the impedance-measurement motivation, duty-cycled operation, and MCU connectivity. This revision intentionally removes several proposed blocks to match the available team and schedule. It should therefore be treated as the current implementation baseline.

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1 Executive Summary

The revised ASIC is a small, one-channel mixed-signal data-acquisition front end. An external MCU drives a low-frequency bipolar excitation across a soil or plant sensor. The return current enters the ASIC's TIA, which converts it into a voltage. On command, an on-chip 8-bit SAR ADC samples that voltage and stores the digital code. The MCU reads the code through SPI and forwards structured measurements to host-side software.

To recover a response correlated with the excitation, the MCU takes samples during both excitation polarities. Software above the MCU subtracts, averages, calibrates, stores, and visualizes those samples. This provides simple synchronous detection without an on-chip demodulator or integrator.

Table 1: Revised architecture at a glance.

Decision	**Baseline**	**Reason**
Sensor channels	One direct current-sense input	Removes the analog MUX, channel registers, and scanning logic.
Excitation	Generated externally by the MCU/board	Removes the excitation DAC and on-chip frequency dividers.
Analog front end	One fixed-gain TIA	Retains the central low-current circuit challenge with a small interface.
Conversion	8-bit SAR ADC	Provides a genuine mixed-signal backend with manageable resolution.
Digital backend	SAR controller, result/status registers, and SPI	Enough digital content to control and read the ADC without a large system FSM.
Signal processing	Host-side subtraction, averaging, and calibration	Creates a substantial CS workstream while removing on-chip demodulation and DSP.
References	External V_{REFH}, V_{REFL}, and V_{CM}	Removes the on-chip bandgap and simplifies first-silicon debugging.
Clocking	External digital clock; no PLL	Predictable, low-risk clock source for the SAR controller.

2 Scope and Requirements

2.1 Blocks included in the ASIC

- electrode return/current-sense input and appropriate pad protection;
- fixed-gain TIA with feedback resistor R_{F} and capacitor C_{F};
- 8-bit SAR ADC analog core: sampling network/CDAC, comparator, and reference switches;
- 8-bit SAR controller;
- one result register plus small control and status registers;
- SPI slave interface;
- reset, enable, data-ready, and basic bias-shutdown control;
- analog test access for the TIA output and ADC input path, subject to pad budget.

2.2 Functions assigned to the MCU and board

- generate the sensor excitation waveform;
- define excitation amplitude and frequency;
- provide the ASIC clock and external reference/common-mode voltages;
- align conversion commands with excitation polarity;
- read ADC codes over SPI;
- package raw samples with excitation state, status, and timing metadata;
- expose a stable measurement interface to the host-side software.

2.3 Functions assigned to the CS software layer

The three CS team members work above the SPI and MCU-firmware boundary. Their system receives structured measurement records rather than manipulating ASIC registers directly. Their work is not limited to transport or visualization. It forms the measurement-intelligence layer that will:

- build a digital twin of the sensor, TIA, ADC quantization, noise, and timing chain;
- design multi-frequency and multi-phase measurement protocols through a host-facing MCU API;
- adapt frequency, amplitude, phase, and repetition count to improve information per unit time or energy;
- recover in-phase and quadrature response with digital lock-in processing;
- estimate complex impedance and fit physically meaningful equivalent-circuit parameters;
- model drift, electrode variation, calibration uncertainty, and out-of-distribution data;
- learn and validate mappings from calibrated electrical features to agricultural quantities;
- produce traceable datasets and quantitative comparisons against fixed measurement protocols.

2.4 Explicitly removed from this revision

The chip does not include a four-channel MUX, excitation DAC, waveform generator, synchronous analog demodulator, low-pass integrator, programmable gain, automatic channel scanning, on-chip averaging, threshold detector, bandgap, PLL, radio, or wireless protocol. These features are not required to demonstrate the revised signal path and would create disproportionate integration risk.

2.5 Requirements and open parameters

Table 2: Requirements and closure method.

Item	Status	Direction	Closure method
Channel count	Fixed	One	Direct electrode-return input; no MUX.
ADC resolution	Fixed	8 bits	Static/dynamic simulation and code-density testing.
Excitation	External	Bipolar, low frequency	MCU/board measurement script; amplitude and frequency remain application parameters.

Item	Status	Direction	Closure method
TIA gain	Open circuit value	One fixed R_F	Select from measured sensor-current range and ADC input span.
Sensor load	Proposal baseline	Two-electrode or interdigitated sensor	Validate first with precision R, C, and RC networks.
Active current	Proposal stretch target	Below 100 µA where feasible	Post-layout block-current budget under a defined conversion rate.
Sleep current	Proposal stretch target	Below 1 µA where feasible	Leakage simulation including pads and always-on digital.
Interface	Fixed	SPI slave plus external clock	MCU timing simulation and board-level transaction test.
Supply/process	PDK dependent	Core and I/O domains per selected process	Use only qualified devices, pads, and ESD structures.

3 System Architecture

3.1 Top-level organization

Figure 1 separates the measurement path from the control path. The sensor drive does not pass through the ASIC. Only the sensor return current enters the TIA. External references set the analog common mode and ADC conversion range.

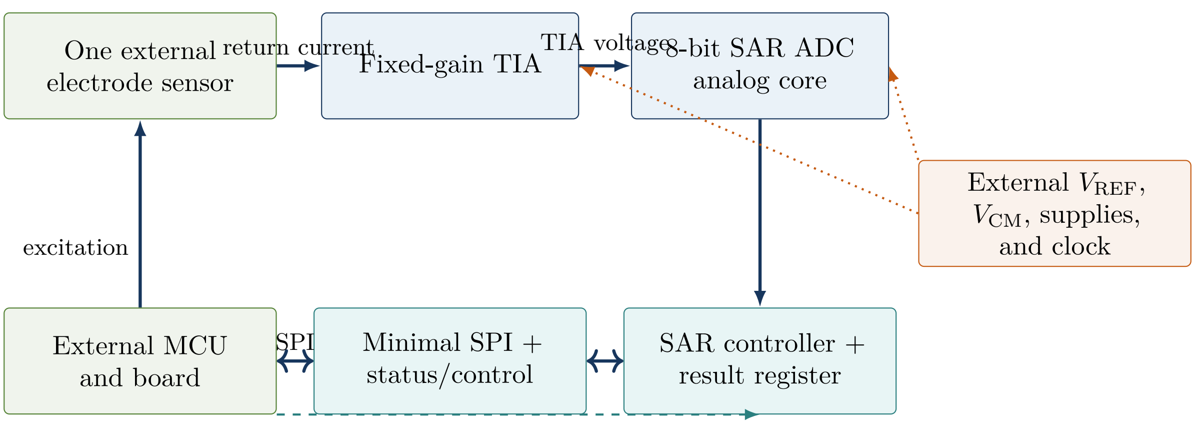

Figure 1: Revised one-channel architecture. The ASIC contains the TIA, ADC, and small digital backend; excitation and signal processing remain external.

3.2 Four paths through the system

1. **Excitation path:** MCU/board → driven sensor electrode.
2. **Measurement path:** sensor return current → TIA → 8-bit SAR ADC → result register.
3. **Control path:** MCU ↔ SPI → start/status/result; external clock → SAR controller.
4. **Software path:** MCU measurement API → adaptive experiment planning → digital lock-in and system identification → agricultural inference.

This organization is intentionally asymmetric. The ASIC receives the difficult low-current analog signal but leaves flexible waveform generation and numerical processing to equipment that is easy to modify after fabrication.

3.3 Above-MCU software architecture

The CS work begins at a stable host-facing measurement API. The MCU and ASIC team are responsible for translating low-level SPI behavior into records containing raw code, excitation polarity, status, configuration, and timing. The CS team uses those records to close a loop: propose a measurement, acquire it, estimate the sensor state, quantify uncertainty, and choose the next most informative measurement. Figure 2 shows that research pipeline.

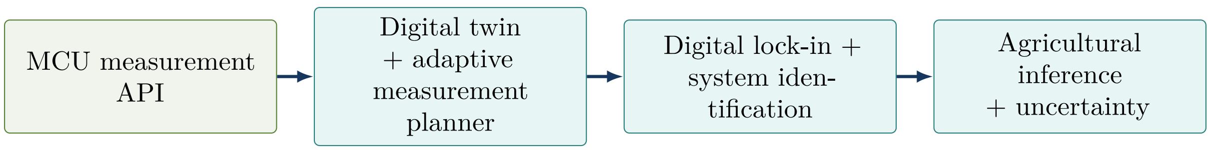

Figure 2: Research pipeline above the MCU boundary. The three blue-green blocks form substantial, connected CS workstreams and can initially be developed against a shared digital twin.

A minimal host record should contain a timestamp or sequence number, excitation frequency, amplitude and phase, raw ADC code, busy/error/overrange status, timing, and sensor/experiment identifier. The host API should accept a measurement plan containing frequency, amplitude, phase offsets, repetition count, and settling time. Freezing these schemas early allows algorithm development to proceed before silicon or the final board is available.

3.4 CS technical contribution and evaluation

The software layer should be evaluated as an engineering contribution, not as interface glue. Three core problems define that contribution.

3.4.1 Digital lock-in and complex impedance estimation

For phase-coherent samples $d[n]$ taken over an integer number of cycles, software estimates in-phase and quadrature components with

$$X(f) = \frac{2}{N} \sum_{n=0}^{N-1} d[n] \cos(2\pi f n / f_s), \tag{1}$$

$$Y(f) = \frac{2}{N} \sum_{n=0}^{N-1} d[n] \sin(2\pi f n / f_s). \tag{2}$$

After gain and phase calibration, (X, Y) produce a complex impedance estimate. The team should compare this estimator with the simpler two-phase subtraction baseline and quantify bias, noise, frequency error, and the number of samples required.

3.4.2 Equivalent-circuit system identification

Known RC loads and sensor sweeps are fit to a selected equivalent-circuit model $Z_{\mathrm{model}}(f;\boldsymbol{\theta})$. Parameter estimation solves

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \sum_k w_k \left|Z_{\mathrm{meas}}(f_k) - Z_{\mathrm{model}}(f_k;\boldsymbol{\theta})\right|^2 . \tag{3}$$

Outputs include fitted resistance/capacitance parameters, residuals, confidence intervals, and a model-adequacy flag. This creates an interpretable bridge between ADC codes and sensor physics.

3.4.3 Adaptive measurement and agricultural inference

A fixed frequency sweep is the baseline. An adaptive planner uses the current parameter estimate and uncertainty to choose the next frequency, excitation amplitude, phase set, or repetition count. The objective is to obtain comparable estimation error with fewer conversions, less energy, or shorter test time. Calibrated circuit parameters and spectral features then feed an agricultural model whose validation must separate training conditions from new days, electrodes, soil samples, or plants.

Success metrics include impedance error on unseen RC standards, equivalent-circuit parameter error, uncertainty coverage, measurement-count or energy reduction relative to a fixed sweep, repeatability across sessions, and generalization error on held-out agricultural samples. A dashboard may present these results, but it is not itself the technical contribution.

4 Measurement Method

4.1 Electrical model

For a sensor impedance $Z_{\mathrm{S}}(j\omega)$ and applied excitation $V_{\mathrm{EXC}}(j\omega)$,

$$I_{\mathrm{S}}(j\omega) = \frac{V_{\mathrm{EXC}}(j\omega)}{Z_{\mathrm{S}}(j\omega)}. \tag{4}$$

The TIA converts this current to a voltage around V_{CM}. The ADC returns samples of that voltage; the ASIC does not directly output impedance, moisture, or conductivity.

4.2 Software synchronous detection

The MCU applies two opposite excitation levels and samples after each half-cycle has settled. Let D_+ and D_- be the ADC codes obtained during positive and negative excitation. The MCU forwards the sample pair, and the above-MCU processing software forms

$$D_{\mathrm{SYNC}} = \frac{D_+ - D_-}{2}. \tag{5}$$

This subtraction rejects much of the static TIA/ADC offset and retains the response that reverses with excitation. Repeating the pair M times gives

$$\overline{D}_{\text{SYNC}} = \frac{1}{M}\sum_{k=1}^{M}\frac{D_{+,k} - D_{-,k}}{2}. \tag{6}$$

The resulting code is phase-sensitive and impedance-related. Calibration with known loads is still required before converting it to an impedance or agricultural quantity.

4.3 Measurement sequence

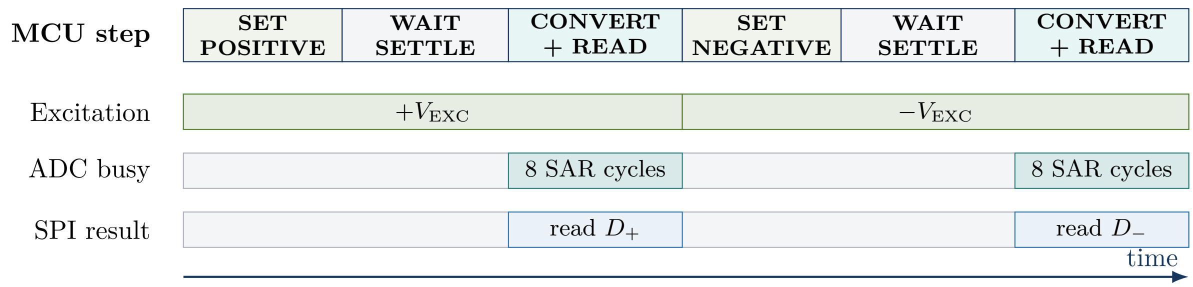

Figure 3: One MCU-controlled sample pair. The ASIC converts individual samples, the MCU packages them, and the above-MCU software computes D_{SYNC}.

The wait interval must be long enough for electrode, TIA, and board transients to settle. This interval is controlled in firmware, allowing it to change after real sensor measurements without modifying the ASIC.

5 Transimpedance Amplifier

5.1 Function and transfer relation

The TIA maintains the current-sense input near V_{CM} and converts sensor current to voltage. For an ideal resistive feedback path,

$$V_{\text{TIA}}(t) = V_{\text{CM}} - I_{\text{S}}(t)R_{\text{F}}. \tag{7}$$

A feedback capacitor stabilizes the loop and limits high-frequency noise:

$$Z_{\text{F}}(s) = \frac{R_{\text{F}}}{1 + sR_{\text{F}}C_{\text{F}}}. \tag{8}$$

The fixed gain should be selected only after bounding the expected sensor current. If the maximum peak current is $I_{\text{PK,max}}$ and the available linear swing is V_{swing},

$$I_{\text{PK,max}}R_{\text{F}} \le 0.8V_{\text{swing}} \tag{9}$$

is a reasonable first margin rule. The remaining headroom accommodates offset, variation, and switching transients.

5.2 Design priorities

The TIA is the principal analog risk. Its design must address:

- stability across the expected sensor and pad capacitance;
- input-referred noise over the MCU's effective measurement bandwidth;
- output range compatible with V_{REFL} and V_{REFH};
- settling after an excitation transition;
- offset and bias current;
- recovery from sensor overload or disconnection;
- enable/disable behavior and sleep leakage;
- pad and ESD leakage relative to the minimum sensed current.

5.3 Fixed gain versus test flexibility

Normal operation uses one feedback network. If schedule and area permit, a test-only alternate feedback resistor may be included to reduce first-silicon risk. It should not expand into a broad programmable-gain feature or require firmware-visible autoranging.

6 8-Bit SAR ADC

6.1 Analog core

The ADC uses a sampling CDAC, comparator, external references, and switch network. During sample, the CDAC acquires the TIA output. During conversion, the SAR controller tests one bit at a time from most significant to least significant. Eight decisions produce one result.

For input range $[V_{\text{REFL}}, V_{\text{REFH}}]$, the ideal code is

$$D = \text{clip}\left(\left\lfloor 256\frac{V_{\text{TIA}} - V_{\text{REFL}}}{V_{\text{REFH}} - V_{\text{REFL}}}\right\rfloor, 0, 255\right). \tag{10}$$

The ideal LSB size is

$$V_{\text{LSB}} = \frac{V_{\text{REFH}} - V_{\text{REFL}}}{256}. \tag{11}$$

Eight bits reduce matching, comparator-noise, and digital-control requirements compared with the earlier 10–12 bit concept. The ADC should still be designed so its error is not the dominant limit after TIA noise and sensor variability are included.

6.2 Analog/digital interface

The analog core receives `sample`, CDAC switch controls, and enable. It returns `comparator_out`. The controller must allow adequate CDAC settling and comparator regeneration time for every bit trial. The external reference pins require local decoupling and a board source capable of handling CDAC switching transients.

6.3 ADC verification

Block verification includes offset, gain error, DNL, INL, missing codes, comparator noise, CDAC matching, reference loading, input kickback, conversion time, and energy per conversion. The ADC must also be testable independently of the TIA through a normally disconnected analog test path.

7 Minimal Digital Backend

7.1 Included logic

The backend contains only:

- an 8-state/bit SAR controller plus sample and completion states;
- an 8-bit SAR trial register;
- an 8-bit result register;
- small control/status registers;
- SPI receive/transmit shift logic;
- reset, busy, data-ready, and enable control.

No SRAM, DSP, channel scheduler, frequency divider, accumulator, or autonomous measurement FSM is needed.

7.2 Conversion state sequence

1. **IDLE:** ADC disabled or waiting; previous result retained.
2. **SAMPLE:** enable the ADC and acquire the TIA output.
3. **BIT7 through BIT0:** set the trial bit, wait for settling, sample the comparator, and keep or clear the bit.
4. **STORE:** copy the SAR register to the result register.
5. **DONE:** clear busy and assert data ready.

Any new start request while busy should be ignored and set an error/status bit. Reset must abort a conversion, disable ADC switching, clear status, and return to IDLE.

7.3 Proposed register map

Table 3: Minimal 8-bit register map.

Address	Name	Access	Purpose
`0x00`	`ID`	R	Fixed design/revision identifier.
`0x01`	`CONTROL`	R/W	Start conversion, ADC enable, soft reset, and test-mode control.
`0x02`	`STATUS`	R/W1C	Busy, data ready, overrange, and command-while-busy flags.
`0x03`	`RESULT`	R	Latest 8-bit ADC result.
`0x04`	`DEBUG`	R	SAR state and selected internal status for bring-up.

SPI mode, edge convention, maximum SCLK, and reset values must be frozen with the MCU test setup. `MISO` is high impedance whenever chip select is inactive.

8 External References, Power, and Pins

8.1 External analog references

The ASIC receives V_{REFH}, V_{REFL}, and V_{CM} from the board. This removes bandgap startup, temperature curvature, reference buffering, and calibration risk from first silicon. It also makes the TIA and ADC easier to characterize with laboratory supplies.

The chip still needs small local bias circuits for the TIA and comparator. These should be derived from the external reference or a simple externally programmed bias current, depending on the PDK and available pins. The reference choice must be frozen before transistor-level design.

8.2 Power states

Table 4: Simplified power behavior.

State	SPI/registers	TIA	ADC	Purpose
Sleep	Retained	Bias off	Off	Minimize leakage while preserving the latest result.
Analog settle	On	On	Off	Allow the TIA to settle before conversion.
Convert	On	On	On	Sample and perform eight SAR decisions.
Readout	On	As commanded	Off	Return result over SPI.

Bias shutdown is preferred to complex header/footer power gating for this revision. The MCU controls the measurement duty cycle.

8.3 Conceptual pin groups

- **Analog:** sensor-current input, optional TIA/ADC test input, V_{REFH}, V_{REFL}, V_{CM}, analog supply, and analog ground.
- **Digital:** SPI clock, chip select, MOSI, MISO, external SAR clock, reset, enable, optional data-ready, digital supply, and digital ground.
- **Board-only sensor drive:** the excitation connection goes from the MCU/driver to the sensor and is not generated by the ASIC.

9 Calibration and First-Silicon Test

9.1 Calibration flow

The board is first connected to known resistors and RC networks. For a chosen excitation and sampling phase, the host software records D_+ and D_- and computes D_{SYNC}. A simple offset/scale model can begin with

$$R_{\mathrm{CAL}} = G\left(D_{\mathrm{SYNC}} - D_{\mathrm{OFF}}\right). \tag{12}$$

More sophisticated mapping should be developed only after real soil or plant electrode data are available.

9.2 Required observability

First-silicon bring-up should support:

- direct observation of the TIA output through a test buffer or switched pad;
- standalone ADC testing with a known external voltage;
- continuous TIA-enable mode for noise and settling measurements;
- visibility of SAR busy/data-ready status;
- result-register reads without starting a new conversion;
- safe reset and analog-disable behavior.

Test switches must be normally open and arranged so their leakage and capacitance do not corrupt normal sensing.

9.3 Bring-up order

1. verify supplies, reset, SPI identity, and quiescent current;
2. verify external references and internal bias points;
3. test the standalone ADC transfer curve;
4. test TIA gain, bandwidth, noise, and settling with precision resistors/current injection;
5. connect the TIA to the ADC and verify end-to-end codes;
6. run positive/negative sample pairs on known RC loads;
7. proceed to controlled soil/plant electrode experiments.

10 Verification and Physical Integration

10.1 Block-level verification

Table 5: Required block evidence.

Block	Minimum evidence before top-level integration
TIA	DC operating point, transimpedance, stability across sensor capacitance, integrated noise, offset, settling, output swing, overload recovery, power, and leakage.

Block	Minimum evidence before top-level integration
SAR analog core	Sampling accuracy, CDAC settling/matching, comparator noise/offset, DNL, INL, missing-code check, reference loading, conversion energy, and PVT/mismatch results.
Digital backend	SPI protocol tests, reset, start/busy/done behavior, SAR binary-search correctness, illegal command handling, synthesis, timing, and digital power estimate.
Top level	Known-load end-to-end codes, reference/pad interaction, mixed-signal regression, extracted critical paths, safe power sequencing, DRC, LVS, and required foundry checks.

10.2 Mixed-signal regression

The top-level testbench should sweep sensor resistance/capacitance, TIA current, ADC input code, external reference variation, supply, temperature, process, and conversion timing. Assertions must ensure that the SAR result is written only after eight valid comparator decisions and that reset cannot leave the CDAC or analog biases in an unsafe state.

10.3 Layout guidance

Place the sensor input pad close to the TIA summing node. Keep SPI, clocks, and CDAC switching away from that node. Locate the ADC near the TIA output while separating comparator/CDAC references from noisy digital return paths. Use guard rings, dense substrate contacts, shielded sensitive nets, symmetrical CDAC layout, dummy elements, and local reference decoupling according to the PDK.

Each owner lays out and verifies their own block. Top-level floorplanning begins as soon as block interfaces and approximate dimensions are stable.

11 Six-Person Team Structure

The three-person ASIC team owns all transistor-level, RTL, SPI, and low-level MCU bring-up work. The three CS team members own the software system above the MCU measurement API. This keeps the silicon interface narrow while giving the software team independent, testable deliverables.

11.1 ASIC team

Table 6: Primary ownership and cross-review.

Owner	Primary work	Reviewer	Boundary responsibility
Krishna	TIA, sensor/pad interface, feedback network, analog input test path	Vidhu	Defines sensor-current range, TIA output range, settling time, noise, and ADC drive requirements.
Vidhu	8-bit SAR analog core, CDAC, comparator, reference loading, ADC test path	Krishna	Defines ADC range, sampling load, comparator timing, and the analog interface to the SAR controller.
Taara	SAR controller, SPI, registers, reset/status logic, RTL verification and synthesis	Vidhu	Defines conversion timing, digital controls into the ADC, MCU transaction behavior, and result validity.

11.2 CS systems-software team

Names can be substituted after the three CS members select roles. Until then, the workstreams are defined by interface rather than person.

Table 7: Above-MCU software workstreams.

Workstream	Primary work	Deliverable and boundary
Adaptive sensing and digital twin	Model the sensor, TIA, ADC quantization, noise, drift, saturation, and timing; use the model to choose frequencies, phases, amplitudes, and repetition counts	A validated simulator and adaptive planner that reduce measurement time or energy relative to a fixed sweep while meeting a specified estimation error.
Impedance estimation and system identification	Implement digital lock-in detection, complex-impedance recovery, calibration, equivalent-circuit fitting, residual analysis, and confidence intervals	A tested estimator that recovers known RC standards and reports parameter error, fit residuals, and uncertainty rather than only plotting ADC codes.
Agricultural inference and uncertainty	Convert calibrated spectra and fitted circuit parameters into application features; evaluate prediction, drift sensitivity, out-of-distribution detection, and uncertainty calibration	A reproducible study on held-out samples that distinguishes useful biological or environmental signal from electrode and measurement artifacts.

The CS team does not own SPI RTL, SAR timing, or transistor-level design. Its development boundary is a host-visible MCU API and a versioned measurement-record schema. Acquisition, logging, dataset versioning, and the simulator transport are shared infrastructure supporting all three technical workstreams rather than an entire member's assignment. The simulator must provide the same interface before hardware is ready.

11.3 Shared integration work

- architecture and interface freeze;
- external-reference and pin decisions;
- mixed-signal behavioral model and top-level regression;
- floorplan, supplies/grounds, pad ring, and test strategy;
- DRC/LVS and extracted top-level verification;
- MCU-to-host measurement schema, simulator, calibration loads, bring-up plan, and final report.

Each block or software workstream has one owner and at least one reviewer. Interface changes are accepted only when the affected schematic/model, signal schema, and tests are updated together.

12 Eight-Week Execution Plan

Table 8: ASIC implementation schedule.

Wk.	Milestone	Analog work	Digital/integration work
1	Interface freeze	Bound sensor current/capacitance; choose reference strategy; TIA and ADC behavioral models	Freeze pins, clocking, SPI mode, register map, SAR handshake, and test plan
2	First schematics	TIA DC/AC loop; ADC CDAC/comparator architecture; nominal simulations	SPI/register RTL; SAR controller; self-checking unit tests
3	Nominal function	TIA transient/noise; ADC transfer and timing; establish ranges	RTL feature complete; ADC behavioral integration; synthesis sanity check
4	Midpoint integration	PVT sweeps, settling, reference loading; begin stable block layouts	End-to-end positive/negative sample regression; interface and scope freeze
5	Block signoff	Mismatch/noise refinement; complete TIA and ADC layouts; begin extraction	RTL freeze; timing/power checks; top-level connectivity and pad controls
6	Top-level assembly	Extracted block simulations; analog test routing; reference/ground plan	Mixed-signal regression, gate-level checks, MCU transaction script
7	Closure	Top-level extracted critical paths; resolve analog and physical issues	Full regression; reset/error tests; DRC/LVS/top-connectivity support
8	Release candidate	Final PVT/mismatch evidence, DRC/LVS/ERC/antenna closure	Final timing, reports, archived source, checksums, bring-up documentation

12.1 Parallel CS software plan

The software schedule runs against simulated records first and then swaps in the real MCU transport without changing the processing or data interfaces.

Table 9: Above-MCU software schedule.

Wk.	Milestone	CS software work
1	Scientific contract	Freeze the record schema, units, phase labels, calibration metadata, benchmark loads, train/test splits, and quantitative acceptance metrics.
2	Digital twin	Model sensor impedance, the TIA, quantization, noise, drift, saturation, and timing; generate reproducible synthetic datasets and failure cases.
3	Complex estimator	Implement two-phase digital lock-in detection and complex-impedance recovery; establish error bars and a fixed-sweep baseline.
4	System identification	Fit candidate equivalent circuits, compare model-selection criteria, inspect residuals, and validate on unseen RC standards.
5	Adaptive planner	Select frequency, amplitude, phase, and repetitions from previous results; compare information, error, time, and energy with the fixed sweep.
6	Hardware integration	Replace simulated transport with the real MCU API while preserving tests; quantify simulator-to-hardware mismatch and update calibration.
7	Agricultural inference	Evaluate application features, held-out generalization, drift robustness, out-of-distribution detection, and uncertainty coverage.
8	Quantitative demonstration	Present known-load and sensor results against baselines; archive datasets, models, metrics, software, and reproducibility instructions.

12.2 Weekly decision gates

- **End of Week 1:** no unresolved ownership, pin, voltage-domain, handshake, or host-schema questions.
- **End of Week 3:** behavioral top level returns the correct SPI code, and the software simulator produces the same host record format.
- **End of Week 4:** no new features; fixed gain and 8-bit resolution remain frozen.
- **End of Week 6:** top-level connectivity and floorplan are stable; software can switch from simulated to real MCU transport.
- **End of Week 8:** required physical checks, end-to-end demonstration, and archived hardware/software evidence are complete.

13 Risk Register

Table 10: Primary remaining risks.

Risk	Prob.	Impact	Mitigation	Fallback
Unknown sensor current	High	High	Measure representative RC loads early; leave TIA test access; maintain headroom	Change board excitation amplitude or feedback option.

Risk	Prob.	Impact	Mitigation	Fallback
TIA instability	Medium	High	Include worst-case pad/sensor capacitance; tune C_{F}; verify phase margin	Lower excitation frequency or increase compensation.
ADC/TIA range mismatch	Medium	High	Freeze common-mode/reference plan and behavioral range model in Week 1	Adjust external references or excitation amplitude.
CDAC/reference kickback	Medium	Medium	Local decoupling, sampling isolation, extracted transient checks	Slow SAR clock; strengthen external reference drive.
Digital coupling into TIA	Medium	High	Floorplan separation, quiet sampling phase, shielding and return-path planning	Stop SPI during sampling; slow clocks.
Hardware/software schema drift	Medium	High	Version the host record; run software against the shared simulator in CI	Freeze a minimal record and add fields only compatibly.
Sleep-current miss	Medium	Medium	Leakage budget including pads; explicit bias shutdown	Relax sleep target while reporting measured duty-cycle energy.
Physical closure late	Medium	High	Begin layout in Week 4; run DRC/LVS continuously	Remove optional test routing before core path.

14 What Must Not Be Added Back

The simplified design succeeds only if its boundary remains stable. During the two-month window, the team should not reintroduce multiple channels, an excitation DAC, analog demodulation, programmable gain, an on-chip bandgap, additional ADC resolution, automatic averaging, wireless, or a PLL. Any of these changes would create new analog interfaces and verification scope.

If schedule pressure still develops, simplify in this order:

1. remove optional debug registers;
2. use a dedicated start pin instead of a writable control register;
3. reduce optional analog test routing while retaining one TIA observation point;
4. retain the TIA, 8-bit SAR conversion, one result register, reset safety, and readable output.

15 Conclusion

The revised ASIC is intentionally compact but still meaningfully mixed signal. Its TIA addresses the difficult low-current sensor interface; its 8-bit SAR ADC performs on-chip analog-to-digital conversion; and its digital backend executes the SAR search, stores a result, and communicates through SPI. The external MCU supplies excitation and flexibility, including phase-sensitive subtraction, averaging, calibration, and application interpretation.

This scope produces three clear ASIC ownership areas for Krishna, Vidhu, and Taara, plus three above-MCU CS workstreams in adaptive sensing, impedance system identification, and agricultural inference with calibrated uncertainty. The main success criterion is not feature count. It is a fabricated, observable signal path whose behavior can be explained from electrode current to SPI code, then carried through a reproducible software pipeline to a calibrated, inspectable result.

A Conceptual Internal Signal Contract

Table 11: Internal and top-level control signals.

Signal	Domain	Source → destination	Meaning
`start`	Digital	SPI control → SAR FSM	Begins one sample/conversion when idle.
`sample`	Digital	SAR FSM → ADC core	Connects the CDAC to the TIA output for acquisition.
`dac[7:0]`	Mixed	SAR register → CDAC	Trial code for the current SAR decision.
`comp_out`	Mixed	Comparator → SAR FSM	Comparator result sampled after settling.
`busy`	Digital	SAR FSM → status	High from sample start through result storage.
`data_ready`	Digital	SAR FSM → status/pin	Latest result is valid and stable.
`result[7:0]`	Digital	SAR FSM → result register	Completed conversion code.
`tia_enable`	Digital	Control → TIA bias	Wakes the front end; requires a defined settling delay.
`adc_enable`	Digital	SAR FSM → ADC bias	Enables comparator/CDAC activity only for conversion.

B Pre-Tapeout Checklist

- ☐ Sensor-current and capacitance range are documented with a conservative margin.
- ☐ External reference, common-mode, supply, clock, pad, and ESD assumptions are frozen.
- ☐ TIA passes stability, noise, settling, overload, power, leakage, PVT, mismatch, and extracted checks.
- ☐ ADC passes DNL/INL, missing-code, noise, timing, reference-loading, PVT, mismatch, and extracted checks.
- ☐ Digital backend passes SPI, reset, start/busy/done, binary-search, illegal-command, synthesis, and timing tests.
- ☐ End-to-end regression maps known TIA input currents to expected SPI result codes.
- ☐ Versioned MCU-to-host records match the software simulator, acquisition service, and processing tests.
- ☐ Lock-in estimation and equivalent-circuit fitting meet error and uncertainty targets on unseen known loads.
- ☐ Adaptive measurement is compared quantitatively with a fixed sweep, and agricultural inference is evaluated on held-out data.
- ☐ Analog test paths are observable and isolated during normal operation.
- ☐ Power-up, power-down, abort, and reset states are safe.
- ☐ Pad ring, supply returns, floorplan, DRC, LVS, ERC/antenna, and extracted critical paths are closed.
- ☐ Source, models, scripts, reports, plots, checksums, and bring-up documentation are archived.

References

[1] Senior Design Team, *Mixed-Signal Impedance Sensing ASIC for Battery-Free Agricultural IoT Nodes*, Senior Design Project Proposal, 2026.